

Brigham Young University,
Provo, UT 84602

December 15, 2025

President Anuradha Hebbar
CEO Action for Inclusion and Diversity
SHRM Headquarters
330 John Carlyle Street
Alexandria, VA 22314

Subject: Barriers for Women in STEM Leadership Solutions Recommendation Report

Dear President Anuradha Hebbar:

There has been a glaring lack of representation of women in STEM leadership for years, and it is a problem we can and should address. Enclosed is my report, *Recommendation Report: Barriers for Women in STEM Leadership and Solutions to Increase Diversity Effectively* and I would be honored if you took the time to look over it. I feel that if we work together, that individuals and organizations can move beyond good intentions and create measurable change for women in STEM.

Despite notable advances in women's participation in STEM, research shows that they remain significantly underrepresented in leadership roles, in spite of their propensity towards the more effective transformational leadership style (Eagly et al., 2003). Research by Haegele (2024) and Glass and Cook (2020) reveal persistent structural and cultural challenges such as the "broken rung" in early promotions (I'd argue in the hiring process as well) along with the "glass cliff" phenomenon that positions women in leadership roles at precarious times. These issues, as well as others I discuss in the following report, not only limit individual growth, but weaken innovation and performance across organizations, and are therefore worth investing time to fix (Lee, 2025). Thankfully these are challenges that SHRM is uniquely positioned to course correct.

This report aims to summarize empirical research and solutions to respond to such problems. Proposed key strategies include: (1) improving transparency in promotion processes, (2) enhancing mentorship and sponsorship networks, (3) addressing cultural biases through leadership training, and (4) reinforcing policies that promote inclusive organizational culture.

As the president of CEO Action for Inclusion & Diversity, your leadership in championing these measures will play a pivotal role in ensuring an equitable, innovative, and high-performing workforce nationwide. Thank you for taking the time to review my recommendations.

Sincerely,

Kimberly Williams

Kimberly Williams
Statistics Data Science Undergraduate, Brigham Young University

Enclosure: Recommendation Report: Barriers for Women in STEM Leadership and Solutions to Increase Diversity Effectively

**Recommendation Report: Barriers for Women in STEM Leadership
and Solutions to Increase Diversity Effectively**

Kimberly Williams

Brigham Young University

WRTG 316: Technical Communication

Dr. Marsha Maxwell

December 15, 2025

Executive Summary

Women remain markedly underrepresented in STEM leadership positions, not because of a lack of capability, but because of systemic and cultural barriers embedded within organizational structures. Research across multiple fields identifies several interconnected issues preventing women from advancing in STEM leadership: early promotion disparities, high-risk leadership appointments, exclusions from informal networks, persistent gender expectations, and daily psychological pressures. These challenges undermine women's career trajectories, weaken organizational performance, and reduce long-term innovation capacity.

This report evaluates these barriers and offers four evidence-based recommendations: First, organizations should strengthen transparency in promotion systems, ensuring that criteria and evaluation processes are clearly defined and consistently applied. Second, institutions must invest in both mentorship and sponsorship programs so that women receive support and advocacy across career stages, regardless of leadership status or position. Third, leadership training should directly address cultural biases and emotional intelligence in addition to practicing more effective transformational leadership styles, equipping leaders to create efficient, equitable, and psychologically safe environments. Finally, organizations should adopt and enforce policies that reinforce inclusivity such as flexible work structures, anti-discrimination protections, and accountability measures tied to leadership performance. Together, these recommendations—transparent promotion systems, robust mentorship and sponsorship, bias-aware leadership training, and inclusive organizational policies—provide a pathway for institutions to cultivate equitable leadership pipelines and improve organizational outcomes.

Recommendation Report: Barriers for Women in STEM Leadership and Solutions to Increase Diversity Effectively

The systemic and interpersonal barriers preventing women from attaining and thriving in technical leadership roles often appear too pervasive to meaningfully address. It's an especially daunting task when looking at the demographics of positions in STEM related fields where women make up only 28% of positions, and, even worse, women of color make up less than 2% (George, 2024). The lack of women in STEM overall is bad enough without looking at the differences in workforce vs. leadership demographics. As an example of the current disparity, in the EU, women only make up 7.9% of engineering professorships even though 32% of scientists and engineers are female (Sagebiel, 2018). This report's purpose is to recommend evidence-based, attainable strategies that institutions and organizations can implement to close the gender leadership gap in technical fields, and raise the percent of women in STEM leadership to at least align with their actual proportion in those fields.

This issue carries significant implications for organizational sustainability. The loss of capable leaders reduces company competitiveness in a market, and company cultures that marginalize women risk diminished creativity and employee satisfaction. Research consistently demonstrates that diverse leadership enhances decision-making and innovation (Askew-Breier & Blau, 2025). Historical and modern barriers illustrate the complexity of the problem. Haegele (2024) identifies the "broken rung" as the point on the 'leadership ladder' at which women are promoted less frequently into first-level management, thereby constraining long-term leadership trajectories. Once women do reach positions of authority, they are often pushed to the edge of the "glass cliff," (i.e. high-risk roles with limited institutional support), and those risky positions with high failure likelihoods lead to burnout or turnover (Glass & Cook, 2020). Cultural factors,

such as male-dominated norms and exclusionary workplace networks, also known as “old boys clubs”, (e.g. “golfing with the guys”, “my old friends from the fraternity”, etc.), further inhibit authentic leadership styles and confidence (Hamrita et al., 2023; Sagebiel, 2018). Psychological dynamics, including social identity threat, defined as, “the threat that people experience in situations where they feel devalued on the basis of a social identity”, exacerbate stress and attrition rates among women in STEM leadership (Hall et al., 2018).

This report will first explore these barriers in greater detail, then examine their consequences for individuals and organizations, and finally, offer actionable recommendations for fostering equitable advancement. These recommendations encompass structural reforms, mentorship and sponsorship models, and leadership training initiatives designed to build inclusive organizational cultures and support sustainable gender diversity in technical leadership.

Why Structural and Cultural Barriers Matter

Women’s underrepresentation in technical leadership is often treated as simply an unfortunate trend, but organizations frequently overlook its broader consequences. Furthermore, research shows that gender gaps harm individual careers and damage entire organizational performance, since “this pattern of career mobility represents a significant loss of highly skilled leadership capital for organizations” (Glass & Cook, 2020). Also, because women use transformational leadership styles, the most effective leadership style according to a meta-analysis of 45 studies on leadership styles, more than men do on average, there is direct harm in terms of lack of effective leadership to companies who exclude women from it (Eagly et al., 2003). These patterns persist not because women lack ability or interest, as evidenced by the fact that 8 out of 10 women desire advancement, but because the systems shaping leadership pathways are built around favoring men by giving them more experience in low level leadership

positions, 100 men for every 87 women (Askew-Breier & Blau, 2025). When bias is built into structures, it shapes innovation, culture, and outcomes.

Sang Eun Lee's study (2025) using 2019 FEVS in U.S. federal agencies showed that higher representation of women in leadership roles is associated with statistically significant stronger performance outcomes, visible in psychological empowerment “(higher group: $\beta = .24$, $p < .001$; lower group: $\beta = .06$, $p < .001$)” and managerial empowerment “(higher group: $\beta = .61$, $p < .001$; lower group: $\beta = .58$, $p < .001$)”. This means that promoting women is not simply an equity initiative; it's a strategic move that strengthens organizations from the inside out. Similarly, Askew-Breier and Blau's study (2025) showed that women's ability to navigate gender bias (supported by emotional intelligence) leads to better job satisfaction and more effective decision-making. When organizations create environments where women's leadership is supported rather than undermined, the payoff is collective, not just individual.

These structural barriers are significant because they come with long-term sustainability costs. When promotion systems unintentionally filter women out of leadership early on, organizations lose talented individuals who could contribute to innovation and continuity. Lee's (2025) findings reinforce that representation is closely tied to empowerment and performance, so excluding women is more than a diversity issue, it is a strategic risk. On a more personal level, research by Hamrita et al. (2023) shows that women studying and working in male-dominated STEM environments frequently feel isolated or undervalued, which increases the likelihood of attrition (Hamrita et al., 2023). When women continually face unsupportive environments, even the most determined individuals may choose to leave, taking their skills, and their potential contributions, with them.

It is important to frame specific challenges, like the broken rung and the glass cliff, as business-critical rather than peripheral. Additionally, psychological strain, specifically social identity threat, affects women's day-to-day interactions in STEM workplaces, influencing their performance and willingness to pursue leadership opportunities (Hall et al., 2018). As aforementioned, these barriers weaken organizational performance, drain talent pipelines, and prevent technical fields from reaching their full innovative potential. Therefore, understanding why these patterns matter is the first step toward changing them.

The Broken Rung and the Glass Cliff

The challenges women face in leadership advancement rarely emerge suddenly at the executive level; rather, they accumulate gradually, beginning with the first managerial promotion. This makes the broken rung especially significant. Haegele (2024) identifies the broken rung as the first major point where women fall behind as the first promotion into frontline management is the most influential point in shaping long-term leadership demographics. It determines who enters the pipeline at all. If women are promoted less frequently at this early stage, it creates a snowball effect that limits how many women can advance later, no matter how qualified they are.

The long-term consequences of this early gap are substantial. Missing the first promotion means missing opportunities for supervisory experience, professional visibility, mentorship from senior leaders or LMX, and eligibility for future development programs (Askew-Breier & Blau, 2025). This helps explain why even fields with growing numbers of women at entry levels still see very few women at the top. Lee's (2025) findings on performance and representation underscore this pipeline issue: organizations benefit when women hold leadership roles, so a broken rung doesn't just affect women, it affects organizational outcomes. For women who do

manage to climb into midlevel or higher positions, the glass cliff presents an entirely different challenge. Glass and Cook (2020) describe this phenomenon as a pattern in which women are disproportionately appointed to leadership roles during times of crisis or instability. These are roles where even the most capable leader would struggle, regardless of their gender, due to limited resources, unclear expectations, or unrealistic performance demands. When women inherit systemic problems and the situation deteriorates, their performance is sometimes interpreted as evidence that they were not fit for leadership, reinforcing stereotypes and making future opportunities less accessible.

The broken rung and glass cliff do not occur in a vacuum. They interact with cultural norms that shape who is perceived as leadership material. Hamrita et al. (2023) report that women in STEM often feel excluded from informal networks that form naturally among men in male-dominated environments. These networks aren't just social, they are pipelines for information, opportunities, and sponsorship; unfortunately, women often have less access to these networks or are not fully integrated into them (Sagebiel, 2018). When women do reach leadership positions, this exclusion means they are more likely to face the challenges of high-risk assignments without the support systems that men often take for granted. Together, the broken rung and glass cliff reveal a leadership pipeline with bottlenecks at the beginning and pitfalls at the top, both shaped by structural inequities and cultural norms that must be addressed to create meaningful change.

Cultural and Psychological Dynamics

While structural barriers shape access to leadership roles, cultural and psychological dynamics influence how women experience their workplaces and how sustainable their leadership trajectories feel. Technical fields, especially STEM settings, often have deeply

ingrained masculine norms that subtly signal who “belongs” and who does not. Hamrita et al. (2023) describe how these norms leave many women feeling like outsiders in environments that prioritize independence, competitiveness, and assertiveness, traits stereotypically associated with men or masculinity. When women do not see themselves reflected in leadership or feel that their authentic ways of working are undervalued, their sense of belonging suffers.

Part of this challenge involves informal networks. Networks in academic and technical settings are crucial for collaboration, research partnerships, leadership opportunities, and recognition, but are often opaque and male-dominated (Sagebiel, 2018). Since much of leadership development happens informally (e.g. through mentoring conversations, social gatherings, or spontaneous collaborations) women who are excluded from these spaces face disadvantages that are hidden but significant.

Cultural expectations also place women in double binds. Helgeson (2017) discusses how communal behaviors such as being warm, supportive, or collaborative are socially expected of women, but these traits are often undervalued or even penalized in leadership contexts that reward assertiveness and independence (Helgeson, 2017). On the other hand, when women do demonstrate more agentic behaviors, they risk being labeled abrasive or unlikable. These conflicting expectations mean that women leaders must navigate a narrower range of acceptable behavior than men, adding emotional labor that is often invisible but draining.

Psychological dynamics deepen these challenges. Hall et al. (2018) describe social identity threat as the anxiety or concern that arises when individuals fear being judged through the lens of stereotypes about their social group. In STEM workplaces, women’s experiences of identity threat fluctuate based on daily interactions such as whether they are listened to in meetings, whether their expertise is questioned, or whether they are the only woman in the room.

Over time, these experiences can affect performance, well-being, confidence, and willingness to take on high-visibility assignments that are often prerequisites for leadership.

These cultural and psychological dynamics matter not only because they shape individual experiences but also because they influence organizational outcomes. Askew-Breier and Blau (2025) show that environments where women feel supported in navigating bias (and where emotional intelligence is part of leadership culture) are associated with greater job satisfaction and stronger persistence. When organizations address bias proactively, teams experience higher engagement and lower turnover, which benefits everyone.

Improving Transparency in Promotion Processes

One of the most effective ways organizations can address the broken rung is by increasing transparency in promotion processes. Many gender disparities arise not from explicit discrimination but from ambiguous, informal, or inconsistent promotion criteria that leave more room for bias. When expectations are unclear, decisions often fall back on subjective judgments about who “seems” ready. Unfortunately, judgments shaped by long-standing stereotypes about leadership fit.

Haegel (2024) emphasizes that the first promotion into management is the crucial intervention point, because it shapes the entire leadership pipeline (Haegel, 2024). Organizations should articulate clear, behavior-based criteria at this level so that employees understand exactly what is required. This reduces the influence of stereotypes on decision-making and helps ensure that women are not overlooked for promotions based on assumptions rather than evidence.

Standardizing promotion procedures also plays an important role. When organizations use consistent rubrics, diverse evaluation panels, and documented rationales for decisions, they

reduce the ambiguity that allows bias to operate. Glass and Cook (2020) connect inconsistent processes to the glass cliff, noting that women are more likely to be tapped for leadership during crises precisely because organizations tend to relax normal procedures when under pressure. By applying standard procedures in all contexts, including crisis situations, organizations can ensure that women are considered for leadership roles under fairer, more stable conditions.

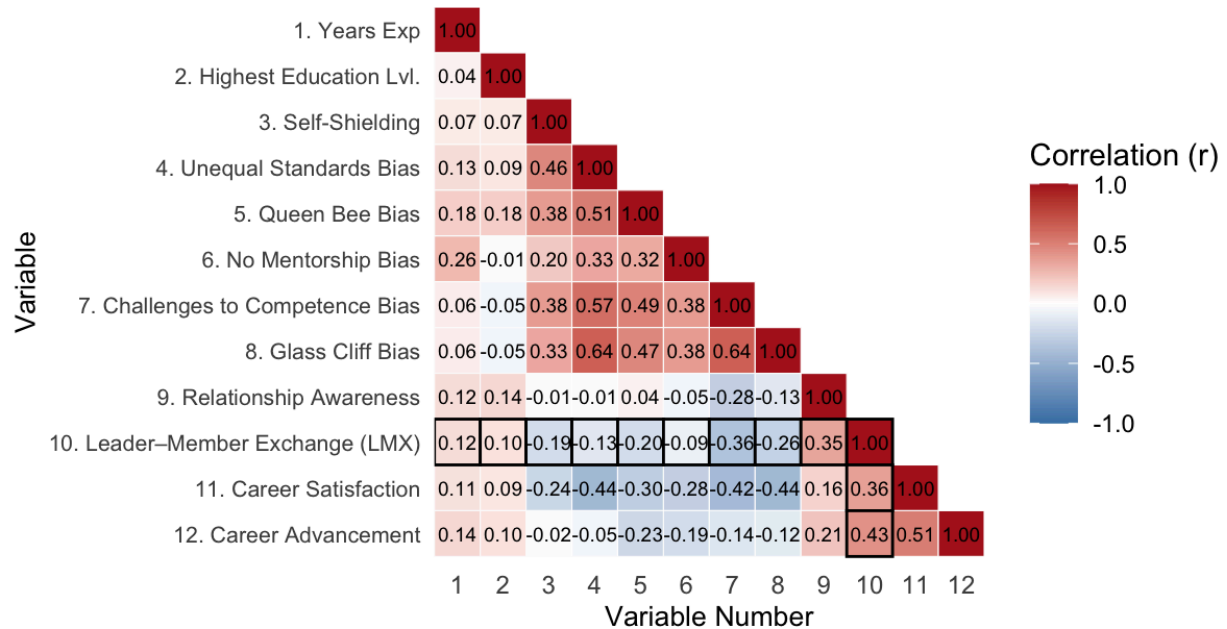
Helgeson's (2017) work also supports the importance of structured evaluations. Because gender stereotypes shape how behaviors are interpreted, structured criteria reduce the risk that women's leadership styles (communal or agentic) are judged unfairly. Another crucial step is regular data collection and analysis. Lee (2025) argues that understanding representation and leadership outcomes requires consistent monitoring, especially because representation itself influences organizational performance. Organizations should conduct equity audits examining promotion rates, performance evaluations, and retention by gender. Askew-Breier and Blau (2025) further support integrating climate and bias assessments into managerial feedback, helping leaders understand how their behaviors shape employees' experiences. Transparency equips organizations with the tools they need to cultivate stronger leadership pipelines.

Enhancing Mentorship and Sponsorship Networks

Network awareness (understanding how networks function and accessing the people within them) is essential for success in academic and technical leadership (Sagebiel, 2018); however, male-dominated networks often operate informally, making it harder for women to enter them without intentional organizational support. Structured mentorship programs, including high quality Leader-Member Exchange (LMX), help people build confidence and navigate unfamiliar environments, as shown in Figure 1, where LMX is significantly correlated with higher career satisfaction and advancement as well as reductions in bias (Askew-Breier &

Figure 1

Correlation Matrix of Variables from Askew-Breier & Blau 2025 study, LMX Highlighted



Note: This figure is a correlation matrix / heat-map based on Table 2 in the paper by Askew-Breier and Blau (2025) which contains further details of the significance of the correlations.

Blau, 2025). Hamrita et al. (2023) show that women in STEM frequently struggle with feelings of isolation, and structured relationships help counteract those experiences by providing community, connection, and guidance. But mentorship alone is not enough. Sponsorship ensures that women are not just supported emotionally but also championed for stretch assignments, leadership programs, and roles that build influence.

Glass and Cook's (2020) work on the glass cliff suggests that strong sponsorship can help women access roles that are not crisis-laden and instead come with real resources and stability. Lee (2025) reinforces that when women hold meaningful decision-making power, organizational performance improves, making sponsorship a strategic investment. Mentorship and sponsorship

relationships can also help women develop skills that improve persistence and leadership success. Askew-Breier and Blau (2025) highlight the value of emotional intelligence in navigating workplace bias and sustaining satisfaction. Helgeson's (2017) work on gender roles can help mentors and mentees understand the double binds women face and strategize how to navigate them effectively. When mentorship and sponsorship programs are intentional, structured, and prioritized, they help correct systemic imbalances and strengthen leadership pipelines.

Addressing Cultural Biases Through Leadership Training

Improving representation and leadership outcomes also requires intentional cultural change, especially through leadership training that addresses bias and stereotype-based expectations. Leaders shape workplace environments, and when they are unaware of how gender norms influence their decisions, even well-intentioned actions can reinforce inequity.

Helgeson (2017) explains how gender stereotypes influence expectations of leadership behavior: men are often assumed to be naturally agentic, while women are judged more harshly for the same behaviors. Because these dynamics influence evaluations, promotions, and day-to-day interactions, bias-aware leadership training should be treated as a core competency rather than a supplemental workshop.

Askew-Breier and Blau (2025) further show that emotional intelligence contributes to more equitable and supportive climates, helping leaders recognize subtle dynamics of bias and respond in ways that improve team satisfaction. Training that helps leaders develop this awareness can meaningfully shift organizational culture.

This training should be grounded in real STEM workplace scenarios. Hamrita et al. (2023) describe common patterns such as women being talked over in meetings or excluded from

informal decision-making that can be directly used in training simulations. Hall et al. (2018) show how micro-interactions influence social identity threat, so training should help leaders recognize how seemingly small behaviors affect psychological safety. By practicing concrete responses to these common situations, leaders become better equipped to create environments where everyone can thrive.

Ongoing reinforcement is essential. Lee (2025) emphasizes that accountability mechanisms such as integrating inclusive behaviors into performance reviews are linked to improved organizational outcomes. Training should not be a one-time event but a continuous part of leadership development.

Reinforcing Policies for Inclusive Organizational Culture

Organizational policies must support and sustain the cultural and structural changes described above. Even the best leadership training cannot compensate for weak institutional protections or policies that inadvertently punish caregivers or marginalized groups.

Strong anti-discrimination and anti-harassment policies, paired with confidential reporting mechanisms and meaningful protections against retaliation, form the foundation of an inclusive culture. Hamrita et al. (2023) show that hostile or dismissive climates in STEM reinforce women's feelings of exclusion and vulnerability, making these policies essential for retention and well-being. Flexible and equitable work policies also play a crucial role. Because gender norms often place disproportionate caregiving responsibilities on women, Helgeson (2017) argues that policies related to flexible work, parental leave, and caregiver support are necessary to ensure women can maintain leadership trajectories without penalty. These policies should be framed not as perks but as structural supports for leadership continuity.

Finally, organizations should tie progress on diversity and inclusion to measurable outcomes. Lee (2025) demonstrates that women's leadership representation correlates with organizational performance, making it logical to integrate diversity metrics into leadership evaluations and recognition systems. Askew-Breier and Blau (2025) add that climates where leaders actively mitigate bias see higher satisfaction and less turnover. When leaders are held accountable for building inclusive cultures, organizational behaviors begin to shift in sustainable ways.

Conclusion

Women's underrepresentation in STEM leadership is not an inevitable trend. It is the result of structural and cultural systems that can be redesigned with intention. By improving promotion transparency, strengthening mentorship and sponsorship networks, offering training that addresses bias and emotional intelligence, and reinforcing inclusive organizational policies, institutions can create environments where women are supported, recognized, and empowered to lead. Doing so benefits not only individual employees, but the entire organization and the broader STEM landscape. As the research shows, when women have equitable access to leadership opportunities, organizational performance improves, innovation increases, and workplace cultures become more resilient and effective.

We have the power to create meaningful mentorship programs, facilitate bias-aware candidate selection, and encourage environments that celebrate the strengths of men *and* women in STEM leadership. Eliminating sexist practices will improve the technical world, not only for women in leadership, but for every company, employee, and student in STEM.

Resources

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Sagebiel, F. (2018). Gender and network awareness in/for successful leadership in academic science and engineering. *International Journal of Gender, Science and Technology*, 10(1), 24–51. <https://genderandset.open.ac.uk/index.php/genderandset/article/view/519>

Word Count: 3987

AI Disclosure: I used AI to write variations of the outline so I could get ideas, used it as a learning/comprehension tool at the beginning of the project to summarize the resources for me, and, at the very end, I used it to fix the wording on about 5 of the sentences (e.g. Throwing away → eliminating) and find phrases to cut out in order to get under the word count; the em dashes however, are all mine.